

Renewable Energy Supply Chain & Logistics Performance: Data Analysis Report

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Tools & Stack: Microsoft Excel

Repository / Dashboard Link:

<https://github.com/fasibor/Renewable-Energy-Supply-Chain-Logistics-Performance>

Executive Summary

The primary question was where logistics delays, material damage, supplier performance gaps, and cost overruns are concentrated, and what can be done to improve delivery performance.

The analysis shows that delays and cost overruns are concentrated on specific routes, while damage rates vary considerably by material and supplier performance remains relatively close..

- **Core Finding:** Logistics performance varies significantly by route and material. The highest delays are concentrated on Rivers → Niger, Oyo → Anambra, and Edo → Kwara, while Delta → Rivers has the highest cost overrun. Damage is highest for Monitoring Equipment (2.23%), while supplier performance scores remain relatively close.
- **Business Impact:** Delays can affect project execution and service availability for host communities, while cost overruns increase freight expenditure. I am targeting a 15% reduction in average delays, 10% reduction in damage rates, and 10% reduction in cost variance within 90 days.
- **Primary Recommendation:** I recommend a 90-day targeted improvement programme focused on the highest-risk routes, stronger handling controls for damage-prone materials, and corrective-action reviews with suppliers and transport partners, supported by monthly KPI monitoring.

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1. Introduction & Background

- **Context:** The analysis focuses on renewable-energy logistics and supply chain operations, where materials must move reliably from suppliers to project locations. Delivery delays, material damage, and freight cost overruns can affect project execution, service availability, and host-community activities.
- **Problem Statement:** The business needs to identify which routes, materials, and suppliers are driving delays, damage, and cost overruns so that corrective actions can be targeted at the areas with the greatest operational impact.
- **Stakeholders:** Management/Executives, Logistics/Supply Chain, Procurement, Operations/Project Teams, Finance, Suppliers, Transport/Logistics Partners, Host Communities, and Community Leaders/Representatives.

2. Project Objectives & KPIs

- **Primary Objective:** Identify the key routes, materials, suppliers, and operational factors driving delivery delays, material damage, and freight cost overruns.
- **Secondary Objectives:**
 - Identify high-delay and high-cost origin–destination routes requiring corrective action.
 - Identify materials with the highest damage rates and potential handling risks.
 - Compare supplier performance based on delivery reliability, damage rates, and cost variance.
 - Examine the relationship between transport distance, transit time, freight cost, and material damage.
 - Identify opportunities to improve logistics efficiency and project delivery reliability.

- **Key Performance Indicators (KPIs):**
 - On-Time Delivery %
 - Average Delay Days
 - Damage Rate
 - Total Freight Cost (₦)
 - Total Cost Variance (₦)
 - Cost Variance %
 - Supplier Performance Score

3. Dataset Overview & Schema

- **Data Source(s):**Excel/CSV mock logistics dataset created for renewable-energy supply chain analysis.
- **Dataset:** 300 shipment records covering renewable-energy logistics activities from January to September 2026, with 25 variables covering suppliers, materials, routes, delivery performance, transit times, damage, and freight costs.

Data Dictionary

Column Name	Data Type	Description	Sample Values
Shipment_ID	STRING	Unique identifier assigned to each shipment	LG-0001
Planned_Date	DATE	Date the shipment was originally scheduled for delivery	08/04/2026
Project_Type	CATEGORICAL	Type of renewable-energy project the shipment supports	Solar Mini-Grid, Commercial Solar
Material_Category	CATEGORICAL	Category of renewable-energy material being transported	Solar Panels, Batteries, Inverters
Supplier	CATEGORICAL	Supplier responsible for providing the materials	SunCore Energy, GreenVolt Nigeria
Origin_City	CATEGORICAL	City from which the shipment originates	Lagos, Port Harcourt, Benin City

Origin_State	CATEGORICAL	State from which the shipment originates	Lagos, Rivers, Edo
Destination_City	CATEGORICAL	City where the shipment is delivered	Abuja, Enugu, Ibadan
Destination_State	CATEGORICAL	State where the shipment is delivered	Niger, Anambra, Kaduna
Priority	CATEGORICAL	Priority level assigned to the shipment	High, Medium, Low
Units_Shipped		Total number of units dispatched by supplier	120
Units_Received	INTEGER	Total number of units received at the destination	98
Units_Damaged	INTEGER	Number of units damaged during transportation	2
Distance_km	INTEGER	Distance travelled from origin to destination in kilometres	300
Planned_Transit_Days	INTEGER	Expected number of days required for delivery	4
Actual_Transit_Days		Actual number of days taken to complete delivery	7
Delay_Days	INTEGER	Number of days the shipment exceeded the planned delivery date	3
Freight_Cost_NGN	FLOAT	Actual cost of transporting the shipment in Nigerian Naira	485000
Budgeted_Freight_NGN	FLOAT	Budgeted or expected freight cost for the shipment	450000
Delivery_Status	CATEGORICAL	Final delivery of shipment	On Time, Delayed
On_Time_Flag	INTEGER	Indicates whether the shipment arrived on	1 = On Time, 0 = Late

		time	
Damage_Rate	FLOAT	Percentage of shipped units that were damaged	2%
Cost_Variance_NGN	FLOAT	Difference between actual freight cost and budgeted freight cost	35000
Cost_Variance_%	FLOAT	Percentage difference between actual and budgeted freight cost	7.3%

4. Methodology & Architecture

Data Extraction: The analysis began with an Excel-based mock renewable-energy logistics dataset containing 300 shipment records and 25 variables, covering shipment details, suppliers, routes, materials, delivery performance, damage and freight costs from 10 January to 6 September 2026.

Cleaning & Validation: The dataset was reviewed for missing values, duplicates, inconsistent data types, invalid dates and calculation errors. Key operational fields such as units shipped, units received, units damaged, planned and actual transit days, freight costs and budgeted costs were validated to ensure the dataset was suitable for analysis.

Feature Transformation: New analytical features were created from the raw data. Route was derived as *Origin State* → *Destination State*, while Distance Bands grouped shipments into ranges such as 0–250 km, 251–500 km, 501–1,000 km and 1,001+ km. Distance Category classified routes as *Short, Medium, Long and Very Long*. Additional measures included Delay Days, Damage Rate, Cost Variance, Cost Variance %, On-Time Delivery and Supplier Performance Score.

Analysis & Insights: The transformed data was analyzed to identify patterns in supplier performance, route delays, distance-related costs, material damage, delivery reliability and freight cost overruns. PivotTables, KPIs, slicers and charts were used to identify high-risk routes, materials and suppliers and translate the findings into actionable logistics improvement recommendations.

5. Data Cleaning & Quality Control

Missing Values

Strategy: Reviewed all 25 fields for blank or null values. No critical shipment records were removed solely because of missing values. Where applicable, records were retained only when the missing field did not affect the calculation of the key logistics KPIs. Critical fields such as Shipment_ID, Planned_Date, Delivery_Date, shipment quantities and freight-cost fields were validated before analysis.

Duplicates

Strategy: Checked for duplicate shipment records using Shipment_ID as the primary identifier and cross-checked shipment attributes such as dates, supplier and route. Duplicate records were removed where the same shipment appeared more than once.

Outliers

Strategy: Reviewed extreme values in Distance_km, Delay_Days, Freight_Cost_NGN, Units_Shipped, Units_Damaged and Cost_Variance_NGN. Legitimate high-distance, high-cost or high-delay shipments were retained because they may represent genuine operational risks. Values that were inconsistent with the underlying shipment records were flagged for validation rather than automatically removed.

Type Conversion & Standardization

Strategy: Standardized Planned_Date and Delivery_Date as date fields and ensured numerical fields such as Distance_km, Units_Shipped, Units_Received, Units_Damaged, transit days and freight costs were stored as numeric values. Categorical fields such as Supplier, Project_Type, Material_Category, Priority, Origin_State and Destination_State were standardized for consistent grouping and PivotTable analysis.

Calculated Field Validation

Strategy: Recalculated and cross-checked Delay_Days, Damage_Rate, On_Time_Flag, Cost_Variance_NGN and Cost_Variance_% against their underlying fields to ensure the derived metrics accurately represented shipment performance.

6. Data Transformation & Feature Engineering

Derived Features

- **Route:** Combined Origin_State and Destination_State to create an origin-to-destination route.
- **Distance_Band:** Grouped Distance_km into predefined ranges: 0–250 km, 251–500 km, 501–1,000 km and 1,001+ km.
- **Distance_Category:** Classified routes as Short, Medium, Long and Very Long based on distance.
- **Delay_Days:** Calculated the difference between Actual_Transit_Days and

Planned_Transit_Days.

- **Damage_Rate:** Calculated the proportion of shipped units that were damaged.
- **Cost_Variance_NGN:** Calculated the difference between actual freight cost and budgeted freight cost.
- **Cost_Variance_%:** Calculated freight cost variance as a percentage of the budgeted cost.
- **On-Time Delivery:** Derived from On_Time_Flag to measure delivery reliability.
- **Supplier Performance Score:** Combined delivery, damage and cost performance into an overall supplier performance measure.
- **Encoding & Scaling:** Categorical variables were retained as categorical fields for Excel PivotTable analysis, filtering and slicers. Numerical variables were kept in their original units because the analysis focused on business KPIs, aggregations and comparisons rather than predictive modelling, so one-hot encoding and numerical scaling were not required.

7. Exploratory Data Analysis (EDA)

Univariate Analysis

- **Distribution Analysis:** Examined the distribution of individual variables to understand shipment distances, delivery delays, material damage, freight costs and cost variances and identify unusually high or low observations.
- **Categorical Distribution:** Reviewed shipment counts across suppliers, material categories, project types, priorities, origin states, destination states and delivery statuses to understand the composition of the dataset.

Bivariate & Multivariate Analysis

- **Correlation Analysis:** Examined relationships between key numerical variables, particularly Distance, Transit Time, Freight Cost, Delay Days and Damage Rate, to identify operational patterns.
- **Supplier Segmentation:** Compared suppliers across on-time delivery, damage rate, cost variance and overall performance score.
- **Route Segmentation:** Analyzed performance across Origin State → Destination State routes, focusing on average delays and freight cost variance.
- **Material Segmentation:** Compared damage rates and shipment volumes across material categories to identify materials with higher handling risks.
- **Distance Segmentation:** Compared logistics performance across Short, Medium, Long and Very Long distance categories and distance bands.
- **Priority & Project Segmentation:** Examined delivery and logistics performance across shipment priorities and project types to identify differences in operational performance.
- **Multivariate Analysis:** Compared multiple factors together; for example, distance, freight cost and transit time—to understand how different operational factors interact and contribute to logistics performance.

8. Key Insights & Findings

Insight 1: Distance Adds Cost and Time, But Not Necessarily Damage

- **Observation:** Longer transportation routes generally require more transit time and higher freight costs, but distance alone does not consistently explain material damage. This suggests that cost and delivery delays may be more directly influenced by route conditions, transportation planning and operational factors than by distance itself.
- **Supporting Metric/Evidence:** The distance analysis showed an upward relationship between Distance_km, Freight_Cost_NGN and Actual_Transit_Days, while damage rates varied considerably across shipments regardless of distance.

Insight 2: A Few Routes Account for the Longest Delays

- **Observation:** Delivery delays are concentrated on a small number of origin-to-destination routes, creating clear areas for logistics investigation and corrective action.
- **Supporting Metric/Evidence:** Rivers → Niger, Oyo → Anambra and Edo → Kwara recorded the highest average delay at approximately 7 days, while Lagos → Delta averaged 6 days.

Insight 3: Cost Overruns Are Concentrated on Specific Routes

- **Observation:** Freight cost overruns are not evenly distributed across the network. A small number of routes account for substantially higher cost variances and therefore warrant closer review of transportation pricing, route planning and carrier performance.
- **Supporting Metric/Evidence:** Delta → Rivers recorded the highest average cost variance at approximately ₦182.4K, followed by Rivers → Kaduna at ₦155.4K and Delta → Edo at ₦130.8K.

Insight 4: Smaller Components Show Higher Damage Risk

- **Observation:** Some smaller and more sensitive renewable-energy components experience higher damage rates than larger structural materials, indicating a potential need for stronger handling and packaging controls.
- **Supporting Metric/Evidence:** Monitoring Equipment recorded the highest average damage rate at 2.23%, followed by Cables & Protection at 2.10% and Hybrid Inverters at 2.07%. Mounting Structures recorded the lowest at 1.56%.

Insight 5: Supplier Performance Is Closely Matched

- **Observation:** Supplier performance scores show relatively little separation, meaning delivery, damage and cost performance are broadly comparable across suppliers rather than being dominated by one supplier.
- **Supporting Metric/Evidence:** SunCore Energy recorded the highest performance score at 94.45, compared with SolarTech Supplies at 91.92, a difference of only 2.53 points.

9. Actionable Recommendations

Short-Term (Immediate Action: 0–30 Days)

1. **Prioritize High-Risk Routes:** Review the Rivers → Niger, Oyo → Anambra and Edo → Kwara routes to identify the causes of the highest delivery delays. Compare transport partners, planned vs. actual transit times, route conditions and shipment schedules, and agree on immediate corrective actions.
2. **Review High-Variance Routes:** Investigate routes with the largest freight cost overruns, particularly Delta → Rivers and Rivers → Kaduna, by reviewing freight rates, route planning and transportation costs.
3. **Strengthen Material Handling:** Introduce additional packaging, loading and handling checks for high-damage materials, particularly Monitoring Equipment, Cables & Protection and Hybrid Inverters.

Medium to Long-Term (Strategic: 1–6 Months)

1. **Route Performance Management:** Establish route-level delivery and cost benchmarks and monitor them monthly to identify recurring bottlenecks and improve route planning.
2. **Supplier Performance Monitoring:** Implement a monthly supplier scorecard covering On-Time Delivery, Damage Rate and Cost Variance, using the existing supplier performance score to track improvement over time.
3. **Logistics Performance Dashboard:** Maintain a centralized dashboard for management, Logistics, Procurement, Operations and Finance to monitor delivery reliability, delays, damage, freight costs and cost variance.
4. **90-Day Improvement Programme:** Set measurable targets of 15% reduction in average delay days, 10% reduction in damage rate and 10% reduction in route-level cost variance, then compare performance against the baseline at the end of the period.

10. Limitations & Next Steps

Limitations

- **Mock Dataset:** The analysis was conducted using a simulated Excel-based logistics dataset rather than actual operational records. The findings therefore demonstrate the analytical approach and potential business questions rather than representing confirmed company performance.
- **Limited Historical Coverage:** The dataset covers approximately **eight months**, which limits the ability to identify longer-term seasonal patterns and year-over-year changes.
- **Limited Operational Variables:** The dataset does not include some factors that could

explain delays or cost changes, such as weather conditions, road conditions, transport partner, vehicle type, fuel prices, loading time, customs/regulatory delays and specific delay reasons.

- **No Direct Root-Cause Data:** The analysis identifies where delays, damage and cost variances occur, but the dataset does not consistently capture the underlying reason for each issue.
- **Potential Sampling Bias:** Because the dataset is simulated, its distributions and relationships may not fully reflect the patterns found in real renewable-energy logistics operations.

Future Scope

- **Predictive Modeling:** Develop models to predict delivery delays, shipment damage and freight cost overruns using historical shipment and operational data.
- **Automated Dashboard Integration:** Connect the logistics dashboard directly to operational databases or cloud data sources so that KPIs and charts update automatically.
- **Expanded Data Sources:** Incorporate transport partner records, GPS/route data, weather information, fuel prices, road conditions and delay reasons to improve root-cause analysis.
- **Real-Time Monitoring:** Introduce shipment tracking and automated alerts for potential delays, high-risk routes, damaged materials and significant cost variances.
- **Longitudinal Analysis:** Expand the dataset across multiple years to identify seasonality, recurring route problems and long-term supplier performance trends.